

Infrastructure Associate



General Engagement with PeopleSERVE:

From roughly April of 2021 through June of 2023, PeopleSERVE resources augmented staffing in the transformation of a 5,000-sf facility. The enclosed outlines the type of technologies and work efforts undertaken by PeopleSERVE. The goal of the below is to help PeopleSERVE staff who worked on this project identify the many technologies they may have assisted with.



Data Center Engagement

Technologies:

➤ Buildout of new Network and Server racks

- Planning, placement, and buildout of Ortronics 45U Network and APC NetShelter 42U Server Racks
- Proper usage of vertical and horizontal cable management
- Documentation of racks including proper labeling or cable connections
- Installation of Dell, Oracle, and Sun servers.
- Installation of Network equipment
- Custom harnessing of low voltage cabling for Telecommunications equipment
- Complete installation of Network, Console, and Power cabling
 - Including but not limited to CAT6, Multi-mode and Single-mode fiber
- Inventory management of Rack Mounts

➤ Data Center Cabling

- Installation of structured cable for CAT 6, 48 port Patch Panels
- Planning and installation of home run cabling for Data Center management

➤ Data Center Power and Temperature Management

- Installation of APC Vertical and Horizontal Metered and switch PDUs
- Application of firmware updates, networking, and custom configuration of APC PDUs
- Installation of Temperature and humidity monitors
- Work within Raised floor plenum to ensure proper airflow.
- Racking or localized cabinet UPS
- Proper placement of devices to ensure proper airflow.

➤ In-Network device management

- Installation and upgrade of Raritan, Dell and BlackBox IP KVMs
- Usage of Crash carts and iDRAC for out of band access to servers

➤ Networking

- Inventory management and installation of Cisco Nexus 9000 series, FEX and SPF modules
- Inventory management racking and cabling of Cisco 4500 series switches.
- Triage of SPF, fiber, and networking issues
- Configuration of DHCP scopes including Address Pool, Scope Options, Reservations, and review of Address Leases.
- Installation of networking devices with requiring different PoE standards including 802.3af (PoE), 802.3at (PoE+) and 802.3bt Type 3(4PPoE)
- Installation of custom, company proprietary PoE equipment



Data Center Engagement

➤ Servers

- Upgrade of Firmware, BIOS and iDRAC to InfoSec standards
- Installation of Windows Server, RedHat Linux and CentOS Linux operating systems
- Configuration of PERC controllers for specific RAID configurations
- Replacement of Hard Drives, installation of memory

➤ Telecommunications Equipment

- Rack and cabling of
 - TDM to IP Media Gateways
 - TDM Channel Banks
 - Cisco 3900 Routers

➤ Configuration and Installation of IPC Trading Turret (Dealer Board)

- Installation of IPC Touch, Pulse, IQMax and Edge Turrets in Performance test Center
- Refurbishment of IPC devices including update of Firmware

➤ Data Destruction

- Leveraged software and tools to properly wipe old media using dBAN software per DOD standards.

➤ E-Waste Management

- Identified and processed older server, network, and telecommunications equipment for proper e-waste recycling.
- Ensured devices were properly sorted and sensitive media was properly disposed of per ISO standards.

➤ Leadership (Kevin)

- Provide instruction to new team members on proper installation and troubleshooting methodology.